

Architecture Selection & Justification

Book Club Reading Challenge & Discussion Portal

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Architecture Selection: We chose Layered Architecture for the Book Club Reading Challenge & Discussion Portal.

Architectural Choice: Layered Architecture. The system is divided into Presentation, Business, and Data layers, matching the structure shown in the component diagram.

Reason 1 – Separation of Responsibilities: The portal has distinct responsibilities such as user interaction, reading goals and progress, chapter discussions and spoiler protection, and book selection and polling. Separating these into presentation and business components keeps each responsibility focused and makes the system easier to maintain.

Reason 2 – Independent Data Management: The Book Club Database is isolated in the Data Layer while the Reading Goals and Progress, Chapter Discussion & Spoiler, and Book Selection & Poll components remain in the Business Layer. This allows the portal's core features to use common data services without directly mixing database operations with business logic.

Security Advantage: Layer separation limits direct access to stored data. Users interact through the User Interface and business components rather than accessing the Book Club Database directly. This provides a clear boundary where access control and validation can be applied before requests reach persistent data.

Performance Benefit: The separation of business functions allows requests to be handled by focused components instead of one large application component. Database access is also centralized through the data layer, which can reduce unnecessary data-access operations and support efficient reuse of stored data.

Conclusion: Layered Architecture is appropriate for the portal because it clearly separates user interaction, application functionality, and persistent storage while supporting maintainability, controlled data access, and efficient processing.